USER MANUAL · 2026

TIKI·100 Emulator

A Norwegian 1984 home computer, running on every modern desktop.

Z80A · 4 MHZ · 64 KB
CP/M COMPATIBLE · TIKO/KP/M BOOT DISK

PRESS F12 FOR MENU
ALT + ENTER FOR FULLSCREEN

The Norwegian computer that taught a generation.

Built around a Zilog Z80A at 4 MHz — the same family that powered the ZX Spectrum and a generation of CP/M business machines.

CPU	Zilog Z80A @ 4 MHz
MEMORY	64 KB main + 32 KB video RAM
GRAPHICS	1024 × 256, 2-colour 512 × 256, 4-colour 256 × 256, 16-colour
SOUND	AY-3-8912 — 3 voices + noise
OS	TIKO/KP/M (CP/M compatible)

WHY IT MATTERED

Adopted by Norwegian schools after winning a major government tender in 1984, putting it on the desks of a generation of pupils.

A FOOTNOTE IN NAMING

Originally called *Kontiki 100*, renamed in 1984 after a naming dispute with the Heyerdahl estate.

An emulator is a program that pretends to be other hardware.

The TIKI-100 emulator runs on a modern PC and behaves as if it were a real TIKI-100 from 1984 — same ROM, same operating system, same software, inside a window on your desktop.

- Full system: CPU, memory, video, sound, two floppy drives, two hard disks, two serial ports, printer.
- Loads original `.dsk` images and runs authentic 1984 software unmodified.
- Modern conveniences — adjustable speed, on-screen menu, save-to-file persistence.
- TCP networking on the emulated serial ports — dial real BBS-style services through a Hayes modem.

Five reasons to keep a 40-year-old machine alive.

Preservation

Real TIKI-100 hardware is rare and ageing. The emulator keeps the platform alive.

Education

Norwegian schoolchildren of the 1980s grew up with this machine — a piece of national computing history.

Software

TIKI-BAS, Tiki-Kalk, BRUM, WordStar, SuperCalc, dBase II, Turbo Pascal — and countless other CP/M classics.

Programming

Z80 assembly, BASIC, Pascal, C — all accessible through period-authentic toolchains.

Curiosity

An afternoon is enough to explore the whole machine from scratch.

Download. Extract. Run.

WINDOWS

Double-click and go.

Download `tiki100-windows-x64.zip` from Releases. Extract. Open `tiki100.exe`.

LINUX · RASPBERRY PI

One-line installer.

Debian, Ubuntu, Mint, Pop!_OS, Raspberry Pi OS — the same script handles them all.

macOS · APPLE SILICON

Brew, then bypass.

SDL2 from Homebrew, then strip Gatekeeper's quarantine flag from the unsigned binary.

```
# keep SDL2.dll next to tiki100.exe
> tiki100.exe
```

```
$ curl -fsSL https://raw.githubusercontent.co
m/HackerCorpLabs/tiki100/main/scripts/instal
l.sh | sh
```

```
$ brew install sdl2
$ xattr -d com.apple.quarantine tiki100
$ ./tiki100
```

Boot the bundled floppy into FD0.

The simplest possible launch — start the emulator with a disk already mounted in floppy drive zero. The machine boots straight into the TIKO/KP/M sign-on.

```
# windows
> tiki100.exe -fd0 disks\boot\tiko_kjerne_v4.01.dsk

# linux / raspberry pi
$ tiki100 -fd0
/usr/share/tiki100/disks/boot/tiko_kjerne_v4.01.dsk

# macOS
$ ./tiki100 -fd0 disks/boot/tiko_kjerne_v4.01.dsk
```

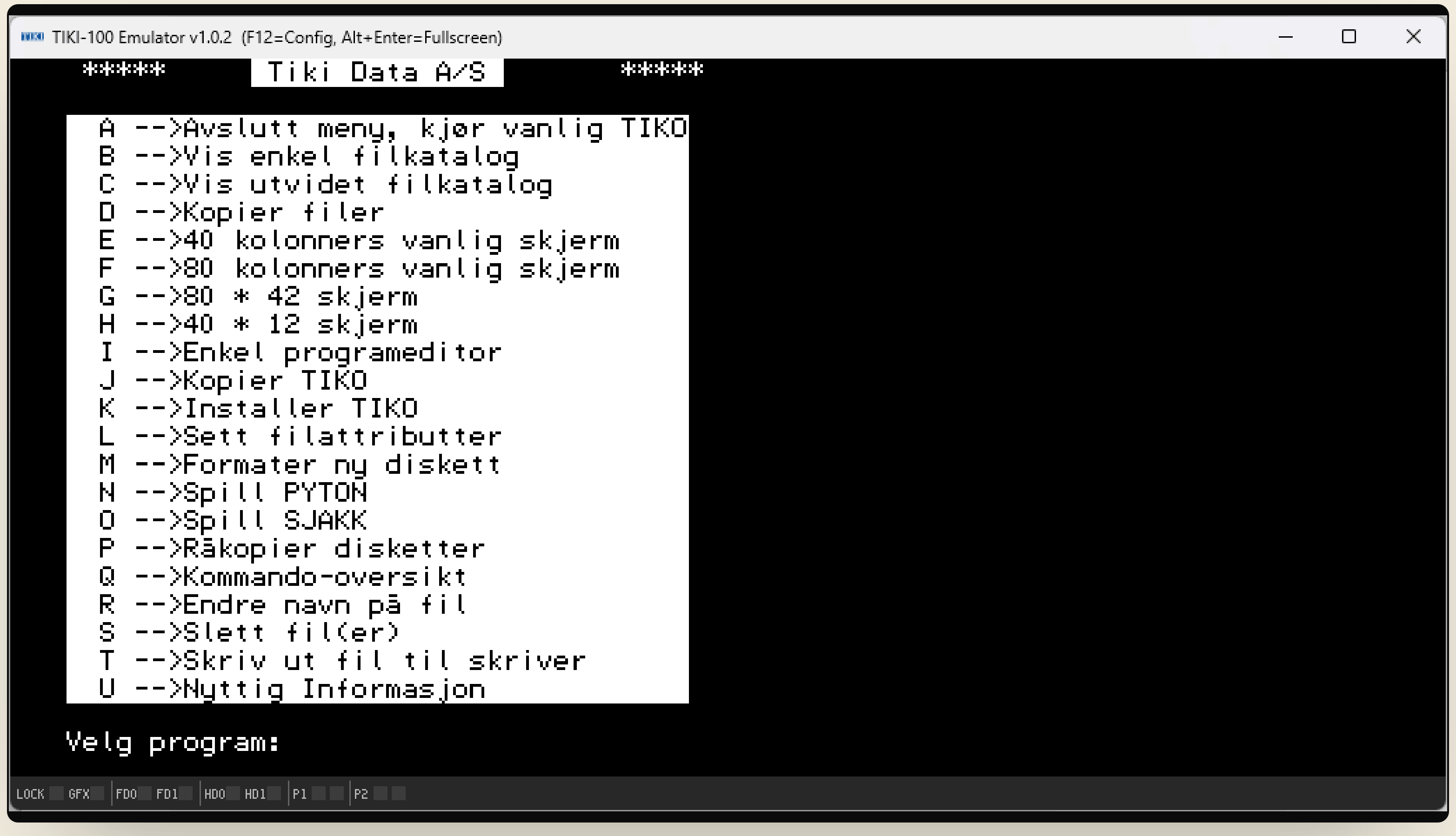


FIG. 01 — TIKO/KP/M boot menu, drive A:

Two regions:
the screen, and the strip.

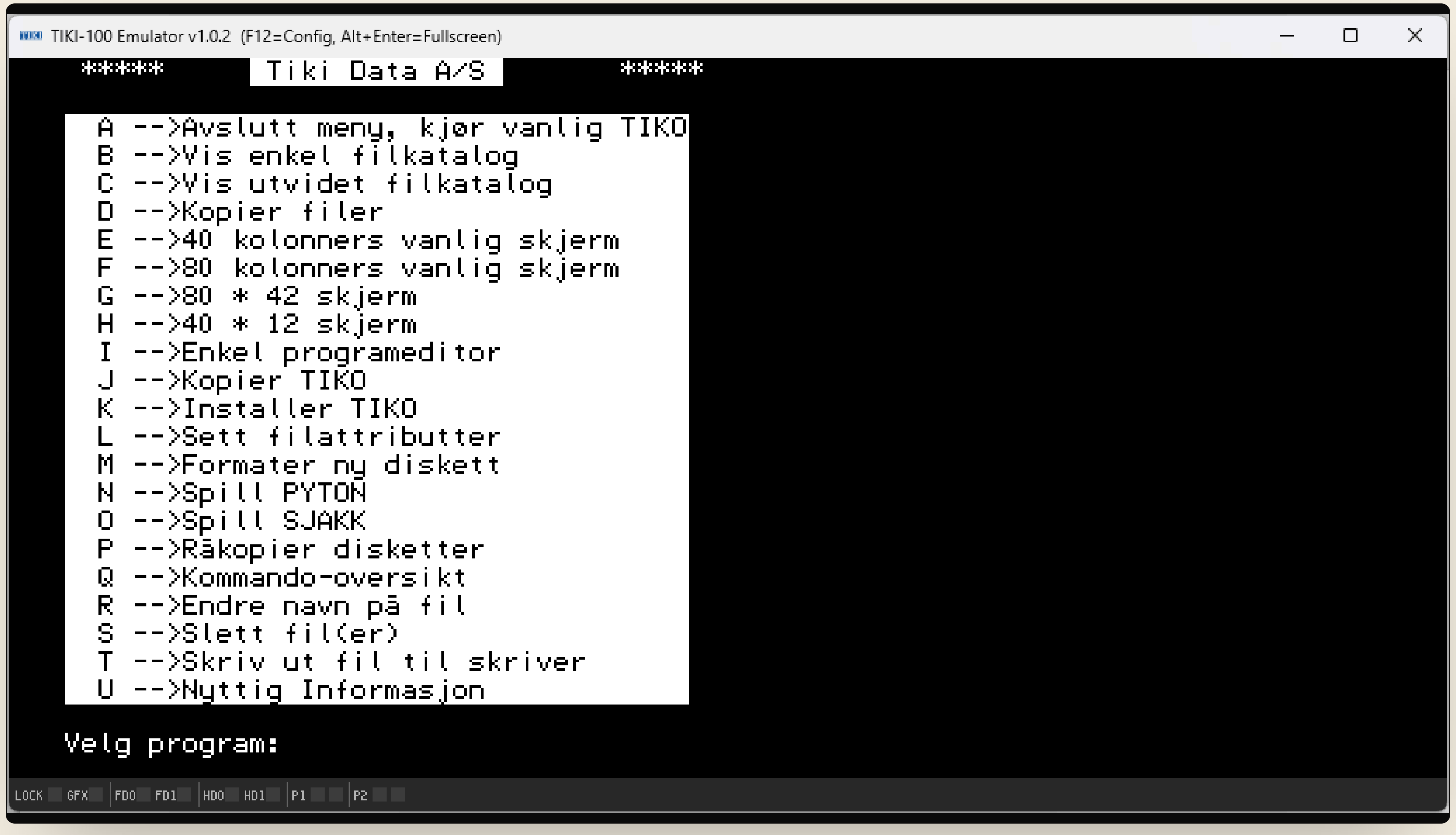


FIG. 02 — Main window: video output (top) and status strip (bottom).

▼ TOP REGION

Video output

The TIKI-100's screen — text mode, game graphics, BASIC prompts, all rendered here at the machine's native resolution.

▼ BOTTOM STRIP

Status bar

Front-panel LEDs that mirror the real machine, plus extras for the emulated serial ports. Hover any LED for a tooltip in the title bar.

flash duration ~150 ms

Six system LEDs, four serial LEDs.

LOCK GFX FD0 FD1 HD0 HD1 | P1·TX P1·RX P2·TX P2·RX

SYSTEM	
LOCK	Caps Lock active
GFX	Graphics mode enabled
FD0 · FD1	Floppy drive selected & active
HD0 · HD1	Hard disk recently accessed

SERIAL	
P1 · TX	Channel A transmitted a byte
P1 · RX	Channel A received a byte
P2 · TX	Channel B transmitted a byte
P2 · RX	Channel B received a byte

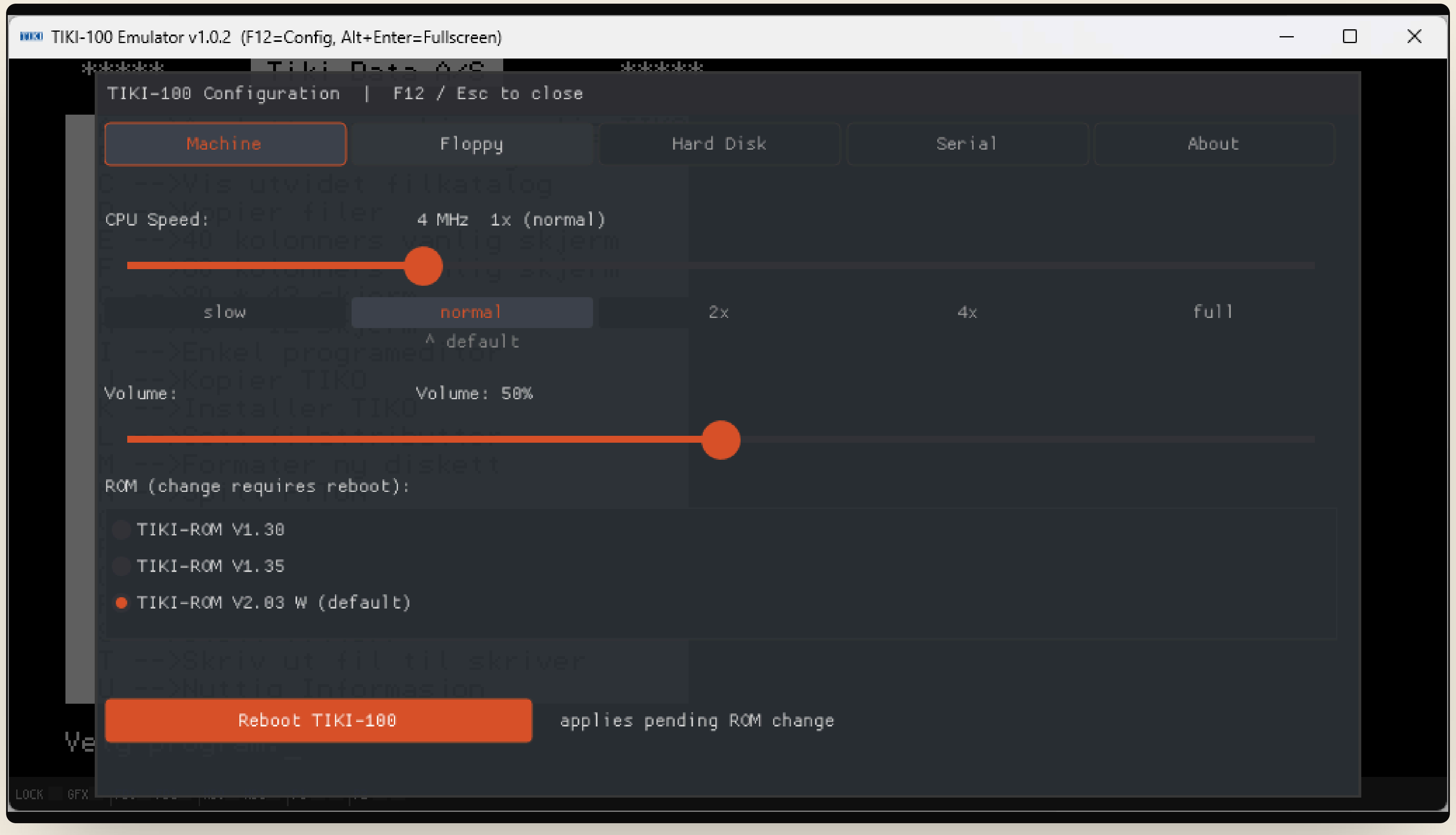
PART · 02

Press F12 for the menu.

The configuration overlay pauses the emulation and exposes five tabs — Machine, Floppy, Hard Disk, Serial, About. Press F12 again, or Esc, to return.

Machine | Floppy | Hard Disk | Serial | About

Speed, sound, ROM, and a soft reset.



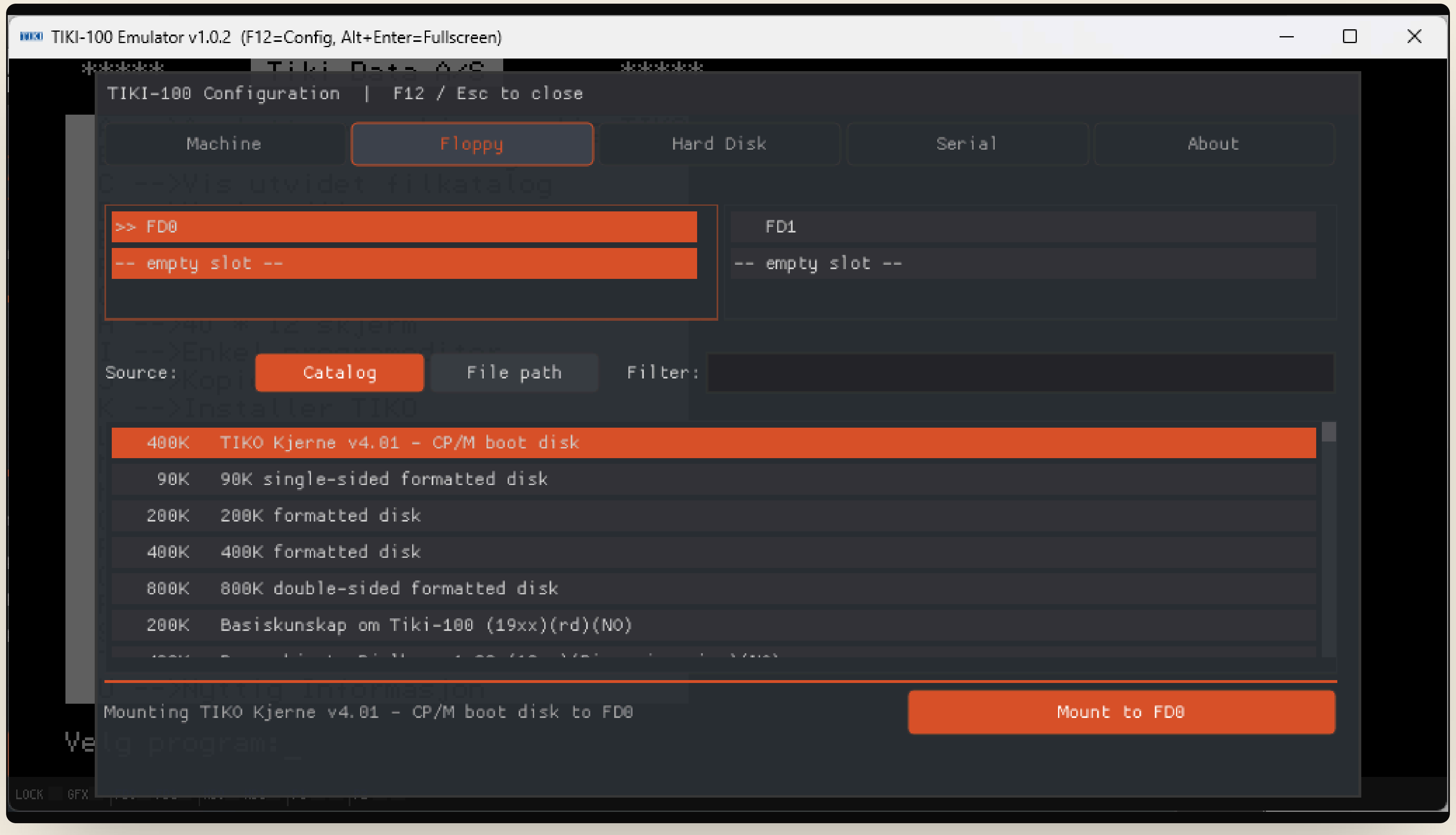
CPU speed

slow	2 MHz
normal	4 MHz · period-accurate
2× · 4×	8 / 16 MHz · long compiles
full	unthrottled · audio may stutter

ROM selection

V1.30 · V1.35 · V2.03 W (default). ROM changes need a soft reboot — handled by the orange button below the picker.

A catalogue of disks. One click to mount.



01 **Browse the catalogue.** Filter by name — type `bas` to narrow to BASIC disks.

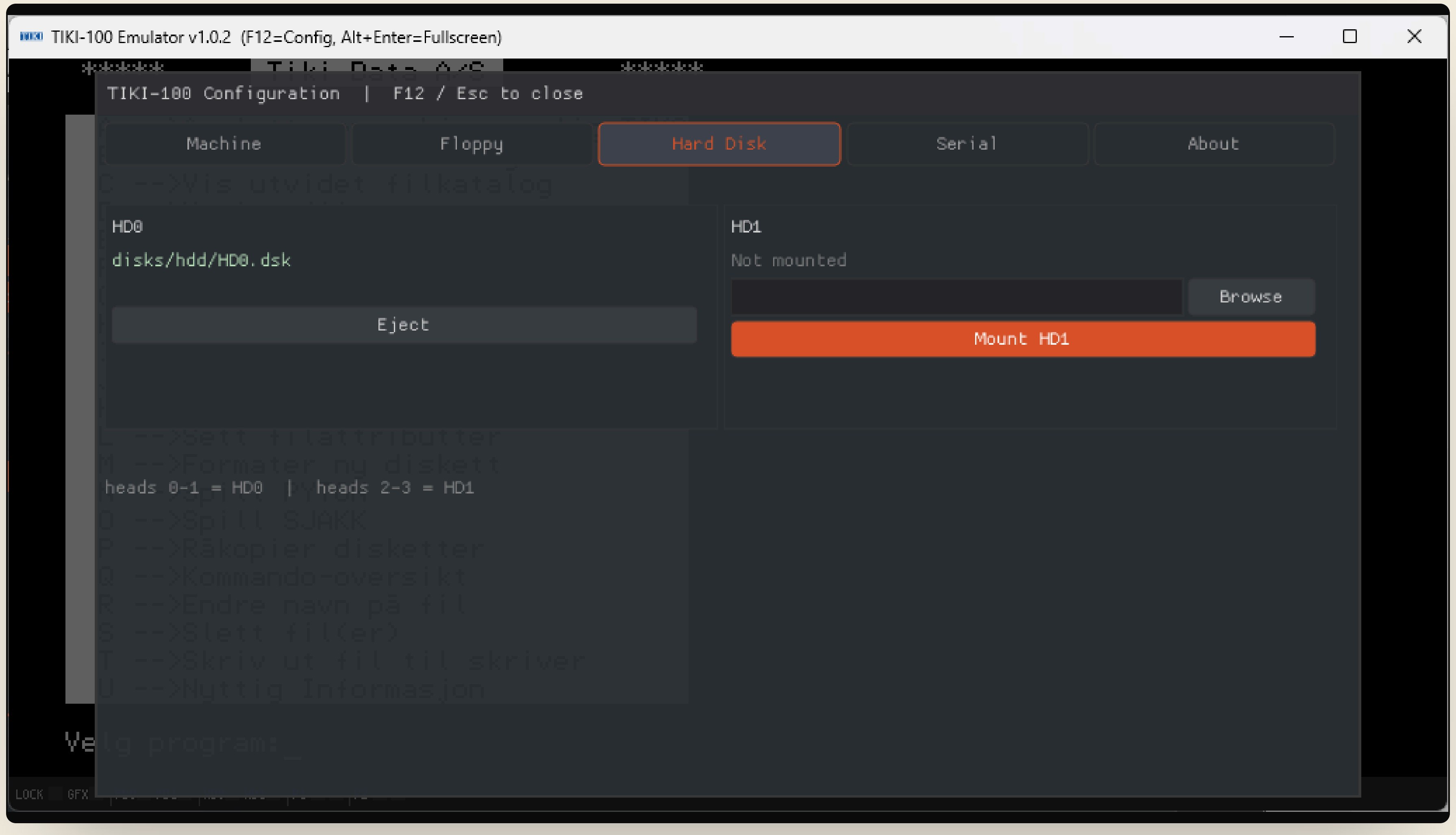
02 **Or pick a file path.** Switch sources to load any `.dsk` from your drive.

03 **Click Mount.** The drive card lights up green when something's loaded; the button switches to *Eject*.

TIP

Copy any `.dsk` you care about before mounting — saves are written back to the file.

Two drives.
Eight megabytes apiece.

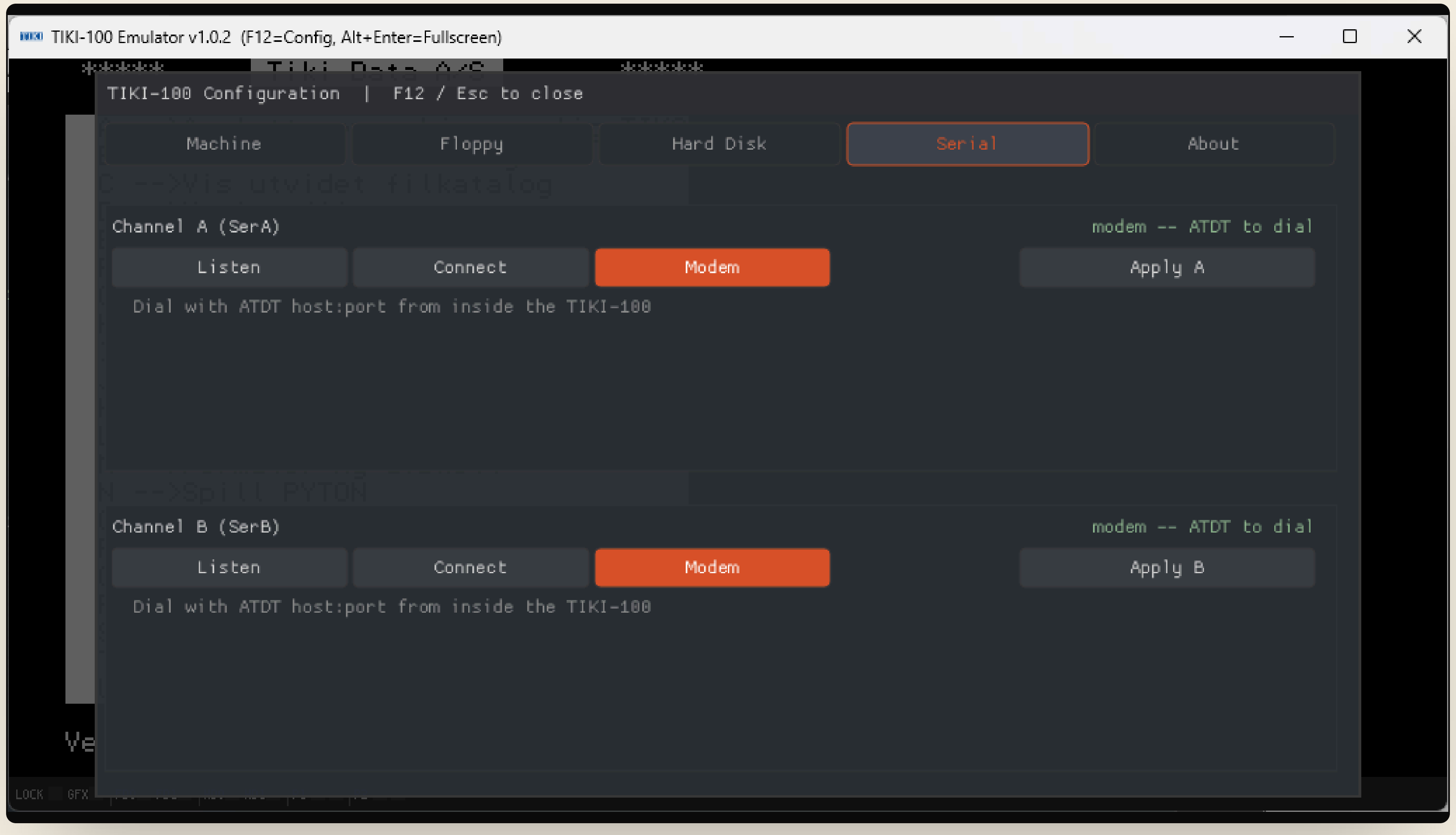


Each hard disk is a single `.dsk` file on your host. Browse to one, mount, and the OS sees it as if it were a real WD1010 drive.

HD0	Heads 0–1 · typically `disks/hdd/HD0.dsk`
HD1	Heads 2–3 · second 8 MB volume

Reset doesn't unmount — disks stay put across a soft reboot. Eject from the drive card to remove.

Two RS-232 channels, three modes each.



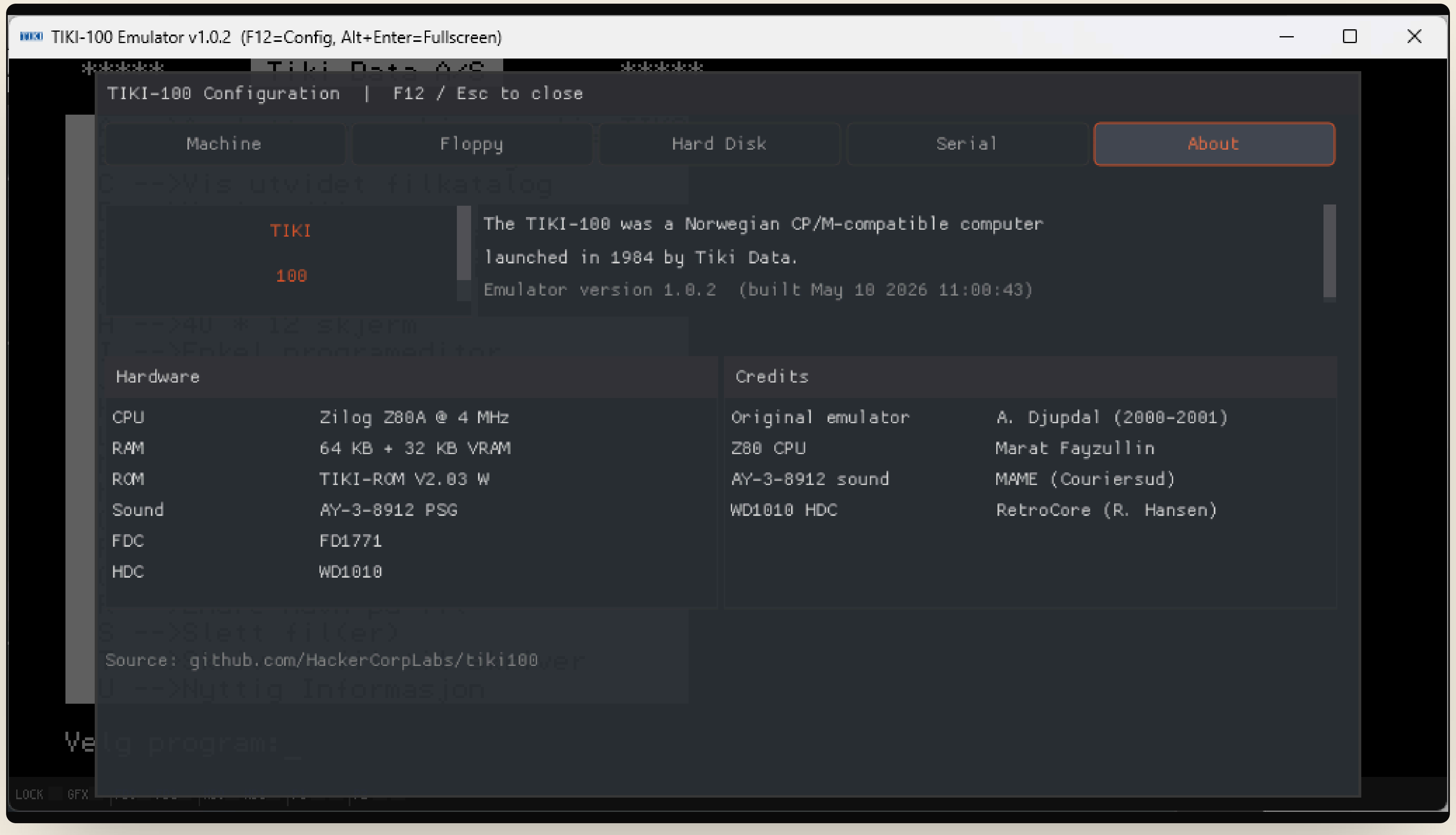
`Modem` — Hayes-compatible. Dial out from inside the TIKI-100 with `ATDT host:port`.

`Listen` — TCP server. Bind a local port; the next inbound connection is wired straight to the DART chip.

`Connect` — TCP client. Permanently linked to a fixed `host:port`, with auto-reconnect.

Modem mode is enough for BBS-style services. Listen and Connect let you wire two emulators together over the network.

The whole machine, on a single card.



CPU	Zilog Z80A · 4 MHz
RAM	64 KB + 32 KB VRAM
ROM	TIKI-ROM V2.03 W
SOUND	AY-3-8912 PSG
FDC	FD1771
HDC	WD1010

CREDITS

Original emulator — A. Djupdal (2000–2001)
Z80 CPU — Marat Fayzullin
AY-3-8912 sound — MAME (Couriersud)
WD1010 HDC — RetroCore (R. Hansen)

Most keys are direct.
A handful are translated.

Translated to TIKI-100 keys

YOUR PC	TIKI-100
Esc	ESC (originally CTRL + Æ)
F10 · Pause	BRYT — break / interrupt
Backspace · Del	SLETT — delete character
F8	GRAFIKK — toggle graphics

Handled by the emulator

SHORTCUT	ACTION
F12	Open / close configuration overlay
Esc	Close overlay (when open)
Alt + Enter	Toggle borderless fullscreen

> The original TIKI-100 had no Esc key — the manual documented CTRL + Æ as the escape sequence. The emulator translates Esc into that combination automatically.

What ran on it.

NATIVE TIKI-100

OS **TIKO/KP/M** — Norwegian CP/M variant. Boots from `tiko_kjerne_v4.01.dsk`.

BAS **TIKI-BAS** — Native BASIC interpreter.

EDU **BRUM** — School learning environment.

CALC **Tiki-Kalk** — Native spreadsheet.

CP/M LIBRARY

DOC **WordStar** — the dominant CP/M word processor.

CALC **SuperCalc** — VisiCalc's CP/M counterpart.

DB **dBase II** — relational database.

DEV **Turbo Pascal 3.0**, MBASIC, FORTRAN, COBOL, M80/L80 assembler.

FIG. 09 · GAME · FULLSCREEN

Forty years on, the games still play.

ALT + ENTER TO TOGGLE FULLSCREEN

Four steps. Five minutes.

01
Boot

Launch with the bundled boot floppy. The OS prompt appears.

```
tiki100 -fd0 boot.dsk
```

02
Mount

Press F12 → Floppy. Pick a disk from the catalogue, click Mount.

```
F12 ▸ Floppy ▸ Mount
```

03
Run

Close the menu, type the program name at the prompt, press Enter.

```
A>> WS ⏎
```

04
Save

Save inside the running OS. Changes are written back to the .dsk file.

```
SAVE "FILENAME"
```

Where to go from here.

Reference docs

- INSTALL.md
- USAGE.md
- MENU.md
- SERIAL.md
- HARDWARE.md

Source & releases

Code, issues, and pre-built binaries live on GitHub.

github.com/HackerCorpLabs/tiki100

github.com/HackerCorpLabs/tiki100/releases

Acknowledgements

Built on the work of A. Djupdal (original emulator, 2000–2001), Marat Fayzullin (Z80), the MAME team (AY-3-8912), and RetroCore (WD1010).

Norsk Teknisk Museum keeps period documentation on the original machine.